Track 2: Visarjan Ghat Watch

Participant guide · Claude Impact Lab × Cerebras Café Compute, Bhopal · Sunday 11 October 2026 · SSR Global Skills Park

Turn three years of buried water tests into a warning Bhopal can act on before Durga visarjan.

Today you build a working tool for a real Bhopal problem, in one day, with Claude and Claude Code. Everything you build is open source and offered to the Government of Madhya Pradesh through MPSeDC.

This guide puts everything for Track 2 in one place: the problem, the data, the rules and how you are judged.

All files are in the Drive folder and on GitHub (same files): https://github.com/Laaaaksh/bhopal-impact-lab-2026, folder `tracks/02-visarjan-ghat-watch`. To clone: `git clone https://github.com/Laaaaksh/bhopal-impact-lab-2026.git`

Problem statement

Background. Every Ganesh and Durga festival, the MP Pollution Control Board (MPPCB) tests the water at Bhopal's idol-immersion ghats three times: before immersion, during, and one week after. The results go into reports of 90 to 120 pages. Nobody lines them up against the legal limits or compares one year with the next.

What the reports show. The numbers buried in those reports are alarming. At Bairagarh (Sant Hirdaram) kund, dissolved oxygen read 0 mg/l four times across 2023 and 2025. Water at zero oxygen supports no fish life at all, and the bathing-water standard asks for at least 5. Across all Bhopal ghat readings from 2023 to 2025, 42 of 116 dissolved-oxygen readings are below 5 mg/l, and 100 of 112 BOD readings are above the bathing limit of 3 mg/l.

Why now. The National Green Tribunal has banned immersion in the Upper and Lower Lakes, and Bhopal Municipal Corporation is building new ghats. Durga visarjan falls on 20 October 2026, nine days after this event.

Who you are building for. The Bhopal Municipal Corporation officer who plans Durga visarjan at the ghats. This officer decides where to send extra cleaning, aeration and water testing. Today that decision rests on memory, because the test results sit in PDFs nobody reads.

Why it is hard.

- The reports print no units. The units in the data are the standard ones for each test.
- A blank means "not sampled", not "clean", and some ghats were sampled only during immersion.
- BDL means below the detection limit: very low, but not zero and not missing.
- Only 4 metals were tested (chromium, lead, zinc, copper).
- One site name, "Rani Kamlapati ghat / Kamla Park kund", may cover two places.

What to build. A tool that gives the officer, before 20 October, a ranked list of Bhopal ghats by risk with the reason for each rank in plain words, the numbers behind every rank traceable to the report and page, and a short brief in Hindi and English the officer can forward. Report the readings against the CPCB limits, say honestly where the data is missing, and don't claim health effects the data doesn't show.

The data

Everything is in the `data/` and `resources/` folders of this track.

Files

File	Rows	What it holds
`data/bhopal_ghats_2023_2025.csv`	1,524	Start here. Every reading at 7 Bhopal ghats, 2023 to 2025, Ganesh and Durga, before / during / after. Spot-checked against the PDFs
`data/sample_bhopal_2025.csv`	453	The 2025 rows of the file above, for quick testing
`data/mp_immersion_sites_2023_2025.csv`	21,600	The same tests at about 100 sites per festival across MP. Less checked. Rows marked `needs_check` have no readable site name
`data/cpcb_water_quality_criteria.csv`	19	The official CPCB water-quality limits, classes A to E
`resources/`	6 PDFs	The original MPPCB reports (2023, 2024, 2025) and Upper / Lower Lake water-quality reports

Columns in the immersion CSVs

Column	Example	Meaning
`year`	`2025`	Report year
`festival`	`Ganesh` or `Durga`	Which immersion the test covers
`site`	`Bairagarh (Sant Hirdaram) kund`	One standard name per Bhopal ghat (Bhopal file only)
`water_body`	`Upper Lake`	Lake or kund the ghat sits on (Bhopal file only)
`regional_office`	`R.O. Bhopal`	MPPCB regional office (statewide file only)
`phase`	`pre`, `during`, `after`	Before immersion, during immersion, one week after
`parameter`	`Dissolved oxygen`	Standard parameter name
`value`	`2.5`	The number. Empty when the report shows no number
`unit_assumed`	`mg/l`	Unit. The reports print no units; these are the standard units for each test
`flag`	`BDL`, `not sampled`	Why `value` is empty
`value_as_printed`	`BDL`	Exactly what the report shows
`parameter_as_printed`	`D.Oxygen`	Parameter name exactly as printed
`site_as_printed`	`Sant Hirdaram Visarjankund, Bairagarh, Bhopal`	Site name exactly as printed
`site_match_confidence`	`high`, `medium`	How sure the standard name is (Bhopal file only)
`needs_check`	`yes`	Site name could not be read (statewide file only)

Column	Example	Meaning
`source_page`	`28`	PDF page in that year's report. Open the PDF and check

The 13 parameters

Parameter	Unit assumed	Plain meaning	Bathing limit (CPCB class B)
pH	none	Acid or alkaline. 7 is neutral	6.5 to 8.5
Dissolved oxygen	mg/l	Oxygen in the water. Fish need it	5 or more
BOD	mg/l	Biochemical oxygen demand: how much organic waste eats oxygen	3 or less
COD	mg/l	Chemical oxygen demand: total pollution that uses oxygen	none set
Total solids	mg/l	Everything left after the water dries	none set
Dissolved solids	mg/l	Solids dissolved in the water	none set
Suspended solids	mg/l	Particles floating in the water. Not in the Durga 2023 and Durga 2025 tables	none set
Turbidity	NTU	Cloudiness	none set
Conductivity	µS/cm	Dissolved salts. Class E caps it at 2,250	none set for B
Chromium, Lead, Zinc, Copper	mg/l	Heavy metals, often from idol paint	none set for B

Codes you will see

Code	Meaning	How to treat it
`BDL`	Below detection limit	Very low. Not zero, not missing
`ND`	Not detected	Very low
`not sampled`	Blank cell, `_` or `–` in the report	Missing. Never treat it as clean
`*`, `**`	Not sampled at that stage	Missing

The 7 Bhopal ghats

Standard name	Water body	Confidence	Note
Bairagarh (Sant Hirdaram) kund	Upper Lake	high	No Durga 2025 table in the report
Khatlapura ghat	Lower Lake	high	
Prempura ghat	Upper Lake	high	
Rani Kamlapati ghat / Kamla Park kund	Lower Lake	medium	Ganesh tables say "Kamla Park kund", Durga tables say "Lower Lake ghat". May be two places
Hathaikheda dam / kund	Hathaikheda dam	high	

Standard name	Water body	Confidence	Note
Malikhedi kund	Kund	high	No Durga 2025 table in the report
Shahpura lake ghat	Shahpura Lake	high	

CPCB water-quality criteria

From `data/cpcb_water_quality_criteria.csv`:

Class	Designated best use	Limits
A	Drinking water source without conventional treatment but after disinfection	Total coliform 50 MPN/100 ml or less · pH 6.5 to 8.5 · Dissolved oxygen 6 mg/l or more · BOD (5 days, 20°C) 2 mg/l or less
B	Outdoor bathing (organised)	Total coliform 500 MPN/100 ml or less · pH 6.5 to 8.5 · Dissolved oxygen 5 mg/l or more · BOD (5 days, 20°C) 3 mg/l or less
C	Drinking water source after conventional treatment and disinfection	Total coliform 5,000 MPN/100 ml or less · pH 6 to 9 · Dissolved oxygen 4 mg/l or more · BOD (5 days, 20°C) 3 mg/l or less
D	Propagation of wildlife and fisheries	pH 6.5 to 8.5 · Dissolved oxygen 4 mg/l or more · Free ammonia (as N) 1.2 mg/l or less
E	Irrigation, industrial cooling, controlled waste disposal	pH 6.0 to 8.5 · Electrical conductivity at 25°C 2,250 micromhos/cm or less · Sodium absorption ratio 26 or less · Boron 2 mg/l or less

Sources

Document	Link	Date
MPPCB Ganesh and Durga Visarjan Report 2025 (92 pages)	https://www.mppcb.mp.gov.in/proc/mon/Ganesh_and_Durga_visarjan_report-2025.pdf	Downloaded 10 Oct 2026
MPPCB Ganesh and Durga Visarjan Report 2024 (118 pages)	https://www.mppcb.mp.gov.in/proc/mon/Ganesh_and_Durga_visarjan_report-2024.pdf	Downloaded 10 Oct 2026
MPPCB Idol Immersion Report 2023, CPCB format (87 pages)	https://www.mppcb.mp.gov.in/proc/mon/Idol_immer._Report_23_CPCB.pdf	Downloaded 10 Oct 2026
Upper and Lower Lake monthly water quality 2025-26	https://xgn.mp.nic.in/File_Upload/MONITORING//Yearly%20Water%20Quality%20Status/2026/July/UPPER%20LAKE-2025-26..pdf	Downloaded 10 Oct 2026
Upper Lake water quality 2023-24	https://www.mppcb.mp.gov.in/proc/mon/Bhopal-2023-24_upper_lake.pdf	Downloaded 10 Oct 2026
Upper Lake water quality by station, 2003-04 to 2022-23	https://www.mppcb.mp.gov.in/proc/mon/comparatives-uppe-lak-lower-lake-2003-2023.pdf	Downloaded 10 Oct 2026
CPCB water quality criteria	https://cpcb.gov.in/water-quality-criteria/	Read 10 Oct 2026
NGT immersion ban and new BMC ghats, Free Press Journal, 26 Feb 2026	https://www.freepressjournal.in/amp/bhopal/mp-news-idol-immersion-to-shift-from-lakes-to-safer-sites-ngt-ban-pushes-bmc-to-build-4-dedicated-ghats	
Durga visarjan date, 20 October 2026, Drik Panchang	https://www.drikpanchang.com/navratri/durga-puja/durga-visarjan-date-time.html?year=2026	

Starting points

These are ideas, not instructions. Pick your own approach.

- Decide which CPCB class fits immersion ghats, and say why. Class B (outdoor bathing) is a reasonable start.
- Your non-technical teammates matter here: researching the NGT order and CPCB guidelines, placing the ghats on a map, writing the Hindi brief, checking numbers against the PDFs.

Constraints

- Public data only. Everything you need is in this folder.
- No connection to any live government system.
- Don't claim health effects the data doesn't show. Report the readings against the limits.

Rules for the day

1. **No pre-work.** Code you submit must be written from 10:15 AM today. Libraries, frameworks and your own general tools are fine.
2. **Use any tools.** Claude, Claude Code, Cerebras inference APIs, or anything else that fits.
3. **Use the data in this pack.** You may add other public data. Name it and link it in your README.
4. **No live government systems.** Don't log in to, scrape, or write to any government portal during the build.
5. **No personal data.** The data here has none. Keep it that way.
6. **One workflow, one user, one outcome.** Build the tool for the officer described above. Don't build a portal.
7. **Teams of 4 to 5,** mixed technical and non-technical. Everyone on the team should have real work.
8. **Your repo must be public on GitHub by 5:00 PM.** The code is offered to the department, so it has to be readable.

How you are judged

The jury includes the organisers and a representative of the problem's real users: a department officer where possible, or a domain expert. Scoring leans toward day-one impact, meaning what the user gets the week after the event.

Submissions close at 5:00 PM. The jury and mentors then review every submission and pick who presents. Not every team presents. Teams chosen to present get 6 minutes to present and demo, then 4 minutes of questions.

Each judge fills one card per presenting team. Every criterion is scored 0 to 5, and day-one impact counts double, so the maximum is 25.

Criterion	What is scored	Scoring guide
Day-one impact (×2)	Is it ready to use, with real impact, on day one?	5 ready, high impact · 4 almost ready, high impact · 3 ready, mild impact · 2 almost ready, mild impact · 1 not ready or not impactful

Criterion	What is scored	Scoring guide
Product	User experience, ease of use, onboarding	5 polished, works first try · 4 well designed, fits the workflow · 3 usable with little learning · 2 functional, needs learning · 1 hard to use
Idea	How creative and impactful the idea is, whatever the result	5 rare insight · 4 creative and original · 3 solid but predictable · 2 close to what exists · 1 already exists
Clarity of demo	Are the problem, the user and the outcome obvious?	5 impeccable pitch and demo · 4 good pitch, convincing demo · 3 ok pitch, demo worked · 2 weak narrative · 1 no narrative or demo failed

For Track 2 the jury looks for:

- The ranking of ghats is traceable to the readings and the CPCB limits.
- Missing data shows as missing, never as clean.
- The brief is short and readable by a BMC officer in Hindi and English.
- The team is honest about what the data doesn't cover.

If two teams tie, the higher day-one impact score wins.

What to submit

Hard deadline: 5:00 PM, Sunday 11 October 2026. Late submissions are not judged.

1. **A public GitHub repo link.** Required.
2. **A screen recording of 2 minutes or less.** Optional, but strongly advised. Record it at 4:00 PM, when the room reminder goes out. A demo that fails live still counts if the recording works.

Submit both through the submission link shared in the group.

Your repo must contain:

Item	Why
`README.md` with team name, members, track, and the one user you built for	The jury reads this first
How to run it, in 5 steps or fewer	The department must be able to run your work after the event
What works and what doesn't	Honesty scores better than a hidden gap
Any extra data you used, with links	Rule 3
A `LICENSE` file	The code is offered to the department. Use MIT unless the organisers say otherwise
No API keys or secrets in the code	Use environment variables. Check before you push

After the event, the organisers connect winning teams with the relevant department officers so the work moves toward real use. Keep your repo public.

Get started in 10 minutes

1. Read the problem statement above.

2. Read "The data" section. It explains BDL, blanks and the units.
3. Open `data/sample_bhopal_2025.csv` and look at five rows together as a team.
4. Agree on your user, your one workflow, and what you will demo.
5. Split the work. Start building.

Questions on the day

Ask a mentor. For anything about the data, check "The data" section of this guide first.

Organisers: Aniket Sahu (Claude Community Ambassador) and Arun Pratap Singh Chandel (Cerebras Community Ambassador).